Bayes' Theorem

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COMP2711 Discrete Mathematical Tools

1 Prior and posterior

Let D mean that a randomly selected person has a disease, and let T mean that the test is positive. We use the same hypothetical model as in the conditional-probability lecture:

$$\Pr[D] = 0.01, \qquad \Pr[T \mid D] = 0.95, \qquad \Pr[T \mid \overline{D}] = 0.10.$$

By total probability,

$$\begin{aligned}\Pr[T] &= \Pr[T \mid D]\Pr[D] + \Pr[T \mid \overline{D}]\Pr[\overline{D}] \\ &= 0.95(0.01) + 0.10(0.99) = 0.1085.\end{aligned}$$

Given a positive result, how likely is the person to have the disease? That is, what is $\Pr[D \mid T]$?

The **prior** is the probability before using this observation: $\Pr[D] = 1\%$. The **posterior** is the probability after conditioning on it: $\Pr[D \mid T]$. These describe our information. Observing the test does not change whether the person has the disease.

Among the positive results.

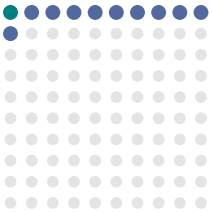
The following table scales the model to 10,000 people. Its entries are expected counts, not guaranteed counts in a random sample.

Group	People	Positive	Negative
Disease	100	95	5
No disease	9,900	990	8,910
Total	10,000	1,085	8,915

Restrict attention to the positive results. The disease fraction is

$$\Pr[D \mid T] = \frac{\Pr[D \text{ and } T]}{\Pr[T]} = \frac{0.0095}{0.1085} = \frac{95}{1085} = \frac{19}{217} \approx 8.76\%.$$

The probability rises from 1% to 8.76%, but remains low. Most people do not have the disease. Even a small error rate in that large group produces more positive results than the disease group does. In particular, $\Pr[T \mid D] = 95\%$ is not $\Pr[D \mid T]$.



Approximate picture of 100 people:

1 positive with disease, 10 positive without disease.

The calculation above uses the exact rates, not rounded dots.

The test is correct overall with probability $0.01(0.95) + 0.99(0.90) = 90.05\%$. Yet a person who tests positive has about 91.24% probability of not having the disease. Overall accuracy and probability of disease after a positive result answer different questions.

2 Bayes' theorem

MUST UNDERSTAND Bayes' theorem updates a prior probability using observed evidence.

For events A, B with $\Pr[A] > 0$ and $\Pr[B] > 0$,

$$\Pr[A \mid B] = \frac{\Pr[B \mid A]\Pr[A]}{\Pr[B]}.$$

Here $\Pr[A]$ is the prior, $\Pr[A \mid B]$ is the posterior, $\Pr[B \mid A]$ is the **likelihood**, the probability of the evidence under A, and $\Pr[B]$ is the overall probability of the evidence.

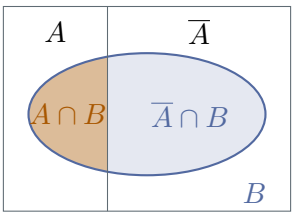
Proof. Conditional probability and the multiplication rule give

$$\Pr[A \mid B] = \frac{\Pr[A \text{ and } B]}{\Pr[B]} = \frac{\Pr[B \mid A]\Pr[A]}{\Pr[B]}.$$

If $\Pr[\overline{A}] > 0$, total probability supplies the denominator:

$$\Pr[B] = \Pr[B \mid A]\Pr[A] + \Pr[B \mid \overline{A}]\Pr[\overline{A}].$$

The numerator is one contribution to the evidence; the denominator includes both contributions. In the diagram, divide the probability in $A \cap B$ by the probability in all of B.



2.1 Several disjoint cases

If $F_1, \ldots, F_m$ are disjoint events covering the sample space, with $\Pr[F_i] > 0$ and $\Pr[T] > 0$, then

$$\Pr[F_j \mid T] = \frac{\Pr[T \mid F_j]\Pr[F_j]}{\sum_{i=1}^{m} \Pr[T \mid F_i]\Pr[F_i]}.$$

This is the same argument: divide the contribution from case j by the contributions from all cases.

3 Inference

Bayes' theorem uses observed evidence to assess possible explanations. We know how likely evidence B is if A is true, and want to infer how likely A is after observing B. For example, after hearing an alarm, how likely is there a fire? After an email mentions a "1 million dollar prize," how likely is it spam? The likelihood alone is not enough: we also need the prior probabilities.

Example: Three factories.

A company receives products from three factories. Let F_A, F_B, F_C mean that a uniformly selected product comes from factory A, B, C, respectively. Let D mean that it is defective.

Factory	Share of products	Defect rate	Defective and from factory
A	50%	2%	$0.50(0.02) = 0.01$
B	25%	4%	$0.25(0.04) = 0.01$
C	25%	8%	$0.25(0.08) = 0.02$

Given a defective product, what is the probability it came from C? First add the contributions from the three disjoint factories:

$$\Pr[D] = \sum_{i \in \{A,B,C\}} \Pr[D \mid F_i] \Pr[F_i] = 0.01 + 0.01 + 0.02 = 0.04.$$

Then apply Bayes' theorem:

$$\Pr[F_C \mid D] = \frac{\Pr[D \mid F_C] \Pr[F_C]}{\Pr[D]} = \frac{0.02}{0.04} = \frac{1}{2}.$$

Factory C supplies a quarter of all products but half of the defective ones.

Problem Set

1. Return to the original medical model, with prior 1%. A person tests negative. Predict whether the disease probability increases or decreases, then calculate $\Pr[D \mid \overline{T}]$.

2. A fair coin selects bag A (three red balls, one blue) or bag B (one red, three blue). Draw one ball uniformly. Given a red ball, what is the probability that bag A was selected?

3. Bags A, B, C are selected with probabilities $0.5, 0.3, 0.2$. Their red-ball probabilities are $1/4, 2/4, 3/4$. Given a red ball, find the probability that each bag was selected.

Solutions

1. A negative result.

The negative-test probabilities are $\Pr[\overline{T} \mid D] = 0.05$ and $\Pr[\overline{T} \mid \overline{D}] = 0.90$. Thus

$$\Pr[\overline{T}] = 0.05(0.01) + 0.90(0.99) = 0.8915, \qquad \Pr[D \mid \overline{T}] = \frac{0.05(0.01)}{0.8915} = \frac{1}{1783} \approx 0.0561\%.$$

The probability decreases from 1%. A negative result is much less likely under disease than under no disease.

2. Two bags.

The red-ball probability is

$$\Pr[R] = \frac{1}{2}\frac{3}{4} + \frac{1}{2}\frac{1}{4} = \frac{1}{2}.$$

Bayes' theorem gives

$$\Pr[A \mid R] = \frac{(3/4)(1/2)}{1/2} = \frac{3}{4}.$$

3. Three bags.

The probabilities of a red ball from A, B, C are respectively $0.5(1/4) = 0.125$, $0.3(2/4) = 0.15$, and $0.2(3/4) = 0.15$. Their sum is 0.425. Hence

$$\Pr[A \mid R] = \frac{5}{17}, \qquad \Pr[B \mid R] = \frac{6}{17}, \qquad \Pr[C \mid R] = \frac{6}{17}.$$

These posterior probabilities sum to one.